Business Intelligence H/W

Knowledge Graph

Note: Answer the following questions in Chinese except that you are a foreign student.

1. Designing and Implementing a Knowledge Graph with Neo4j.

Graph analytics empowers users to analyze and visualize complex relationships between entities using nodes (entities), edges (relationships), and properties (attributes). In this assignment, you will conceptualize and implement a Knowledge Graph using Neo4j, based on a subject area you are personally interested in or have expertise in. The goal is to support decision-making through graph-based queries.

Objectives

(1) Select a Topic of Personal Interest or Expertise
Choose a topic you are passionate about or familiar with (e.g., digital marketing, financial products, supply chain, healthcare procedures, university courses, music industry).
Your KG should include:

- At least 20 distinct nodes (entity types and instances).
- At least 30 edges with meaningful properties (e.g., dates, weights, conditions).

(2) Define and Implement the Graph Schema

- Identify the main entity types (e.g., Person, Topic, Decision, Resource).
- For each type, define several instances.
- Define relationships between instances, with properties that support analysis (e.g., relevance score, time spent, frequency).

(3) Implement the Knowledge Graph in Neo4j

- Use Neo4j Desktop or Neo4j Aura Free for implementation.
- Import node and relationship data using CSV files or CREATE statements.
- Use Cypher queries to define nodes, relationships, and properties.

- Use Neo4j Browser or Neo4j Bloom to visualize your graph clearly, with different shapes or colors for different node types.

(4) Query the Graph for Decision Support

- Write and document at least 5 Cypher queries that demonstrate how the graph can support decision-making.
- Example queries:
  - "Which topics are most central to my interests?"
  - "Which resources are recommended based on co-learning patterns?"
  - "Who can I consult for a topic I want to explore next?"
  - "What decisions have the most dependencies?"
  - "What are the most common learning paths?"

Submission Requirements

(1) Neo4j Project

- Submit your Neo4j Browser project or Cypher script used to create the KG.
- If you used CSV import, include the CSV files for nodes and relationships.

(2) Graph Visualization

- Include a screenshot of your graph visualization from Neo4j Browser or Bloom.
- The graph must clearly distinguish between different node types and relationships.

(3) Cypher Queries

- Submit at least 5 well-commented Cypher queries used for decision-making.
- Each query should include a short explanation (1–2 sentences) about its purpose.

(4) Reflection (Optional Bonus)

- Briefly reflect (150–200 words) on how using graph structure helped you explore or make decisions in your chosen topic.

2. 本題作業旨在培養以知識概念化(knowledge conceptualization)強化敘述性分析學的能力，藉由將本學期 BI 課程所學的知識進行盤點與統整，並以知識圖譜的形式加以視覺化。藉此訓練學子的組織與分析能力，促進知識的共享與創新。

概念化的重要性：知識概念化對企業至關重要，將隱性領域知識概念化並結構化儲存為知識圖譜，不僅提高了搜索效率與決策支持能力，亦促進了知識共享與創新，最終將企業知識資源轉化為可管理與評估的資產，從而增強企業的競爭力和適應性。

作業內容(繳交的內容)：

(1) 請運用心智圖(如下所列之工具與說明)或 infographic (如 piktochart 工具)，將 BI 課所學之知識概念化為知識圖譜(如課堂所展示之案例)。請著重說明各概念之間的關係。

(2) 請以所設計的 KG 或 infographic 為基礎， 提出您在本課程中所獲得關於自己業務乃至個人思維與人生觀的洞見（insight）。

(3) 列出幾項就自己工作或生活相關的專業管理與實務應用的 well-defined problems/opportunities。

心智圖工具:

- https://coggle.it/
- https://tobloef.com/text2mindmap/
- https://www.playpcesor.com/2018/08/text2mindmap.html
- https://deanlife.blog/coggle-toturial/
- https://markmap.js.org/repl
- Xmind